Tomasz Krzeminski

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Education

Northwestern University — Nov 2025 - June 2026
Master of Science in Mechanical Engineering — *Evanston, IL*

Northwestern University | Combined Cumulative GPA: 3.71 — Sept 2022 - June 2026
Bachelor of Science in Mechanical Engineering — *Evanston, IL*

Engineering Experience

HAND ERC — **Evanston, IL**
Research Associate — June 2026 - Present

- Developed a custom automated manufacturing platform for novel electrohydrodynamic fiber actuators from concept to operation in 3 weeks, independently designing and assembling mechanical hardware, sourcing and integrating all electronics, wiring motion-control systems, and developing custom C++ control software
- Engineered synchronized rotary and linear motion-control algorithms to eliminate 100–200 μm winding defects, achieving consistent fiber placement, reliable actuator fusion, and repeatable manufacturing performance through process-control optimization and stage synchronization
- Reduced actuator production cycle time by $>$ 50% (60+ min to ∼30 min) and decreased material preparation time by 73% (30 min to ∼8 min) through automation development, workflow redesign, and manufacturing process optimization
- Increased manufacturing yield from ∼ 20% to ∼ 80% (1 successful actuator in 5 attempts to 4 successful actuators in 5 attempts) by resolving process-control, motion-synchronization, material-handling, winding-stability, and thermal-processing challenges in a novel actuator manufacturing process

Hollister Incorporated — **Libertyville, IL**
Global Engineering Co-Op — June 2025 - Present

- Led the design, fabrication, and validation of a manufacturing testbed used to investigate process failures, executing 8 experimental trials across 4 tooling configurations to evaluate material behavior and support root-cause analysis efforts that informed production corrective actions
- Performed dimensional and flatness studies on production components, identifying 0.001–0.002" surface variation in parts processed within a 0.003 in. process-control window; translated findings into new tooling concepts aimed at improving process robustness and reducing failure rates
- Designed and released CAD models, manufacturing drawings, and custom tooling assemblies supporting process-development and validation initiatives, utilizing tolerance analysis and design-for-manufacturing principles to develop practical engineering solutions for production testing
- Developed a one-dimensional heat-transfer model to estimate process heating requirements and sealing times, using analytical methods to guide tooling modifications and support process-development decisions
- Built and implemented a laboratory-wide 5S management and auditing system spanning 8 operational zones, increasing departmental compliance from 43% to 80% within 6 months through standardized assessments, automated data tracking, and continuous process improvements
- Developing Siemens PLC programming and industrial automation expertise through the upgrade of a manufacturing test platform, creating control software to replicate production operating modes and support efforts to mitigate a multi-million-dollar manufacturing scrap issue
- Engineered, analyzed, manufactured, and deployed robotic end-effector tooling modifications to accommodate product design changes, performing structural and failure-mode analyses to develop a solution meeting 0.200" clearance requirements while minimizing modifications to existing production equipment

Northwestern Baja SAE — **Evanston, IL**
Powertrain Engineer — Sept 2024 - June 2025

- Designed 4 key components of the front differential housing (body, lid, input, and endcap) in SolidWorks, reducing assembly weight by 3% compared to the 2024 design
- Developed 3 Siemens NX CAM programs for a HAAS VF-3 CNC and modified G-code to support in-process inspection, ensuring critical press-fit bearing features met $\pm$0.0025" tolerance requirements
- Machined differential housing components across 10 manufacturing operations and 80+ CNC hours, designing custom

fixtures for multi-sided machining while maintaining critical bearing interfaces within ±0.0025" of nominal dimensions

- Improved differential housing mounting and load paths through enhanced fixturing features, increasing structural support and minimizing deflection from drivetrain forces and moments

Powertrain Engineer Sept 2023 - Jun 2024

- Designed front Hazardous Release of Energy (HROE) powertrain guarding for the 2024 vehicle, reducing manufacturing and assembly costs by 15%
- Developed a serviceable fastening system utilizing plastic rivets, reducing disassembly and maintenance time from 30 minutes to 6 minutes
- Manufactured shafts, spacers, and custom mechanical components using manual milling and turning processes to support drivetrain assembly and testing
- Created parametric SolidWorks parts and assemblies for powertrain guarding and drivetrain components, applying design-for-manufacturing principles to improve manufacturability and assembly efficiency

Midtown Metro Achievement Center **Chicago, IL**

Engineering Instructor Jun 2024 - Dec 2024

- Designed and instructed an engineering fundamentals course for high school students, emphasizing hands-on project work and practical application of concepts
- Created and implemented projects and learning materials that effectively engaged students and strengthened their understanding of Algebra and Physics, leading to a 50% improvement in exam scores

Projects

Fiber Pump Winding Rig Jun 2026 - Present

- Designed and built an automated fiber pump winding rig integrating linear and rotary motion stages, custom CAD assemblies, stepper motor control, and embedded electronics to manufacture electrohydrodynamic microfluidic pumps
- Developed winding control software that synchronized linear and rotary stage motion to achieve target fiber helix geometries, including homing, initialization, and automated winding sequences based on pump design parameters
- Optimized the manufacturing process through experimental validation and iterative design improvements, significantly reducing setup time, improving winding consistency, and increasing production scalability

Robot Finger | Speedster Team Wesbite Mar 2026 - Jun 2026

- Served as Mechanical Design Lead for a 5-person team developing a high-bandwidth robotic finger for Northwestern University's Robot Design Studio capstone project
- Engineered a tendon-driven, 3-DOF finger mechanism featuring custom linkage and transmission systems, achieving 27.9 rad/s joint speed and 12 Hz bandwidth while meeting strict size and force requirements
- Drove CAD design, prototyping, manufacturing, and system integration efforts, contributing to a robotic platform capable of superhuman-speed interaction tasks and real-time trajectory tracking

Sawyer SDK Port (ROS1 to ROS2) | SawyerSDK Mar 2026 - Jun 2026

- Ported the Sawyer Robotics SDK from ROS 1 to ROS 2, enabling modern robot control, motion planning, and development workflows for legacy hardware
- Integrated MoveIt 2 motion planning and trajectory execution capabilities, supporting robot visualization, joint control, and autonomous arm movement
- Developed and tested robot interface components, including gripper control, joint trajectory actions, and robot enablement tools to improve system reliability and usability

Skills

Software: C++, C, Python, MATLAB, Office Suite, LaTeX, Git, OpenCV, Docker, Linux, Siemens PLC

Mechanical: SolidWorks, Siemens NX (CAD & CAM), Onshape, ANSYS, Conversational Mill, Lathe, CNC, 3D Printing

Robotics: Embedded Systems, ROS/ROS 2, MoveIt, Kinematics, Machine Learning, RViz, PF and UKF